IT Service Desk Performance & SLA Analytics Dashboard

Portfolio Project Documentation

Tools: Excel | MySQL | Power Query | Power BI | DAX

Project type: Self-created portfolio project using synthetic/sample IT service desk data

1. Project Overview

Developed an end-to-end IT Service Desk Analytics Dashboard using a sample dataset of 20,000 IT support tickets. The project analyzes ticket volume, resolution performance, SLA compliance, priority distribution, and operational workload.

2. Project Objectives

• Analyze overall ticket volume and status.
• Monitor ticket resolution performance.
• Evaluate SLA compliance across ticket priorities.
• Identify locations and departments with higher ticket volumes.
• Track monthly ticket trends.
• Present insights through a Power BI dashboard.

3. Tools & Technologies

• **Excel:** Initial dataset review and validation.
• **MySQL:** Data validation and SQL analysis.
• **Power Query:** Data transformation and preparation.
• **Power BI:** Dashboard development and visualization.
• **DAX:** KPI calculations and analytical measures.

4. Dataset

The dataset contains 20,000 ticket records covering January 2025 through August 2026. It includes eight locations (Mumbai, Bangalore, Chennai, NSL 1, NSL 2, Mohali, Vizag, and Coimbatore), four departments (IT Helpdesk, IT Network, IT Server, and Desktop Support), four priority levels (P1–P4), ticket statuses, assigned engineers, and resolution information.

5. Data Preparation & Analysis

The dataset was reviewed in Excel for missing values and structure. It was imported into MySQL staging table **ticket_data**, validated, and prepared in **ticket_data_final**. SQL analysis covered record counts, duplicate Ticket IDs, missing values, date ranges, status, priority, location, department, engineer performance, resolution time, and SLA compliance. A view named **vw_it_service_desk_powerbi** was prepared for Power BI reporting. Power Query was used to review data types and prepare reporting fields. Missing resolution values were retained as missing rather than replaced with zero.

6. Key KPIs

Metric	Result
Total Tickets	20,000
Resolved Tickets	15,517
Open Tickets	1,635
Pending Tickets	1,244
On Hold Tickets	1,029
Cancelled Tickets	575
Resolution Rate	77.59%
SLA Compliance	51.87%
Average Resolution Time	35.72 hours

7. SLA Calculation

SLA targets were defined by priority: P1 within 4 hours, P2 within 8 hours, P3 within 24 hours, and P4 within 48 hours. A resolved ticket was counted as meeting SLA when its resolution hours were within the relevant target. Unresolved tickets and resolved tickets with missing resolution hours were excluded. Of 15,425 eligible resolved tickets, 8,001 met SLA and 7,424 breached SLA, resulting in 51.87% compliance.

8. Dashboard Visualizations

The single-page Power BI dashboard contains eight KPI cards and five visualizations: Monthly Ticket Trend; Ticket Distribution by Priority; Location-wise Ticket Volume; SLA Compliance by Priority; and Department-wise Ticket Volume.

9. Key Findings

- Overall resolution rate was 77.59%.
- SLA compliance among eligible resolved tickets was 51.87%.
- P3 had the highest ticket volume, with 9,036 tickets.
- Mumbai had the highest location-wise volume, with 6,302 tickets.
- IT Helpdesk had the highest department-wise volume, with 8,457 tickets.
- SLA compliance varied by priority, from 50.99% for P3 to 60.26% for P1.

10. Project Outcome

Created a consolidated single-page Power BI dashboard to summarize service desk KPIs, monthly trends, workload distribution, resolution performance, and SLA compliance. The dashboard can help stakeholders identify operational patterns and areas that may need further investigation.

11. Dataset Disclaimer

This is a self-created portfolio project using synthetic/sample IT service desk data. It does not contain confidential company, employee, or customer production data.

Prepared as a learning and portfolio demonstration. Results describe this sample dataset and should not be interpreted as actual organizational performance.